

MOBILE RADIO NETWORK

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Planning Optimization Project

Mathematical formulation:

Sets

- T: Set of test points (locations requiring coverage)
- C: Set of candidate locations for installing base stations
- B: Set of base station types = {macro, micro}

Parameters

- $PR[c,t,b]$: Received power (in dBm) from base station type b at site c to test point t
- $THR[c,t,b]$: Achievable throughput (in Mbps) from base station b at site c to test point t
- $cost[c,b]$: Installation cost of base station type b at site c
- γ : Minimum required signal strength (sensitivity threshold in dBm)
- θ : Minimum required throughput per test point (7 Mbps)

Decision Variables

- $x[c,b] \in \{0,1\} \rightarrow 1$ if a base station of type b is installed at candidate site c
- $y[c,t,b] \in \{0,1\} \rightarrow 1$ if test point t is served by a BS of type b from site c

Objective Function

Minimize total network deployment cost:

$$\min \sum_{c \in C} \sum_{b \in B} cost_c^b \cdot x_c^b$$

Coverage Constraint

Each test point must be covered by exactly one base station from any type (macro or micro)

$$\sum_{c \in C} \sum_{b \in B} y_{c,t,b} = 1 \forall t \in T$$

This prevents over- or under-coverage.

Power Constraint

A test point cannot be served by a BS if the received signal is too weak:

$$y_{c,t,b} = 0 \text{ if } PR_{c,t,b} < \gamma$$

Throughput Constraint

Each test point must receive at least 7 Mbps in total from the BS it's connected to:

$$\sum_{c \in C} \sum_{b \in B} THR_{c,t,b} \cdot y_{c,t,b} \geq \theta$$

Link Activation Constraint

A test point can only be served by a BS if that BS is actually installed:

$$yc, t, b \leq xc, b$$

This ties the usage of a BS to its installation.

Results of the planning Scenarios and the answer to the big telco questions.

We tested three deployment strategies: macro-only, micro-only, and mixed-mode. The mixed-mode scenario, combining both macro and micro base stations, resulted in the lowest total cost (4.58) while still meeting all coverage and throughput requirements. This confirms that a hybrid deployment is the most cost-efficient and effective solution.

Scenario	Total Cost (€)
Macro-only	(7.99)
Micro-only	(4.77)
Mixed-mode	(4.58)

Q1. Given the geographic area, can you plan a wireless network deployment that meets the requirements?

Yes, our team of three successfully planned a wireless network deployment that satisfies all the given requirements using a Mixed Integer Linear Programming (MILP) model. We ensured that:

- All test points (TPs) are covered by at least one base station (BS),
- The received signal strength exceeds the required threshold (γ),
- Each TP achieves a peak throughput of at least 7 Mbps,
- And the overall deployment cost is minimized.

We implemented and solved the MILP in AMPL for multiple deployment strategies.

Q2. Is it better to use only macro or only micro base stations in terms of cost?

We evaluated macro-only and micro-only scenarios separately. The results were:

- Micro-only deployments were more cost-effective due to lower installation costs.
- Macro BSs cover larger areas but are significantly more expensive.

Final Costs:

- Macro-only: €7.99
- Micro-only: €4.77

Therefore, micro-only is preferable when only one type must be chosen.

Q3. What is the cost reduction if a mix of macro and micro base stations is installed?

We tested a mixed deployment that allows both macro and micro-BSs at any candidate site. This hybrid model produced the lowest total cost.

Mixed-mode cost: €4.58

Reduction vs macro-only: 42.66%

Reduction vs micro-only: 3.88%

This proves that a combined deployment of macro and micro base stations results in a more cost-efficient and flexible network layout, while still satisfying all performance requirements

BELOW PROVIDED THE LINK FOR COLAB:

https://colab.research.google.com/drive/117whU_TWrP4JlHMZDaCUkZeCcflQ1Uh?usp=sharing